EXP NO:1)A Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib /seaborn for visualization.

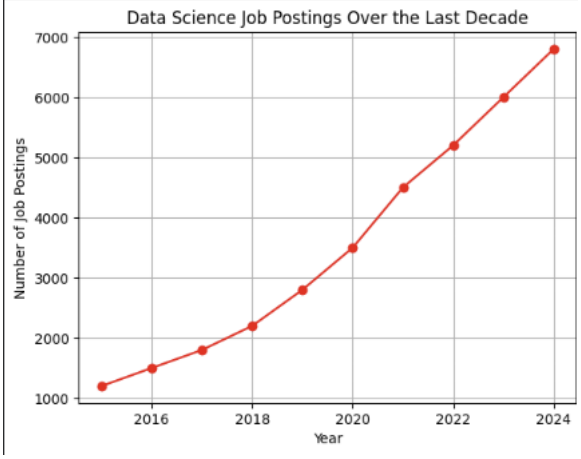
AIM:

To Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib /seaborn for visualization.

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    "Year": [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024],
    "Job_Postings": [1200, 1500, 1800, 2200, 2800, 3500, 4500, 5200, 6000, 6800]
}
df = pd.DataFrame(data)
print(df)
plt.plot(df["Year"], df["Job_Postings"], marker='o',color="red")
plt.title("Data Science Job Postings Over the Last Decade")
plt.xlabel("Year")
plt.ylabel("Number of Job Postings")
plt.grid(True)
plt.show()
```

```
   Year  Job_Postings
0  2015          1200
1  2016          1500
2  2017          1800
3  2018          2200
4  2019          2800
5  2020          3500
6  2021          4500
7  2022          5200
8  2023          6000
9  2024          6800
```



RESULT:

Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib /sea
born for visualization.

EXP NO:1)B Analyze and visualize the distribution of various data science roles (Data

Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

AIM:

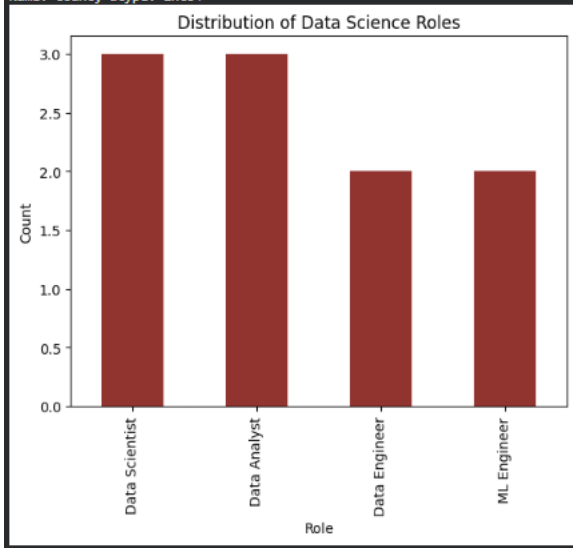
To analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    "Role": ["Data Scientist", "Data Analyst", "Data Engineer",
             "ML Engineer", "Data Analyst", "Data Scientist",
             "Data Engineer", "Data Scientist", "ML Engineer",
             "Data Analyst"]
}
df = pd.DataFrame(data)
role_count = df["Role"].value_counts()
print(role_count)
role_count.plot(kind="bar", color="brown")
plt.title("Distribution of Data Science Roles")
plt.xlabel("Role")
plt.ylabel("Count")
plt.show()
```

OUTPUT:

```
Role
Data Scientist    3
Data Analyst      3
Data Engineer     2
ML Engineer       2
Name: count, dtype: int64
```



RESULT:

Thus the program Analyze and visualize the distribution of various data science roles from a dataset.

EXP NO:1)C Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

AIM:

To Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

PROGRAM:

```
import pandas as pd
structured_data = pd.DataFrame({
    "Name": ["John", "Mary", "Sam"],
    "Age": [20, 25, 22]
})
semi_structured_data = {
    "Name": "John",
    "Age": 20,
    "City": "Chennai"
}
unstructured_data = [
    "Hello, this is a text file",
    "Photo.jpg",
    "Video.mp4"
]
print("Structured Data:")
print(structured_data)
print("\nSemi-Structured Data:")
print(semi_structured_data)
print("\nUnstructured Data:")
print(unstructured_data)
```

OUTPUT:

```
Structured Data:
   Name  Age
0  John   20
1  Mary   25
2   Sam   22
```

Semi-Structured Data:

```
{'Name': 'John', 'Age': 20, 'City': 'Chennai'}
```

Unstructured Data:

```
['Hello, this is a text file', 'Photo.jpg', 'Video.mp4']
```

RESULT:

Thus the program to Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

EXP NO:1)D Conduct an experiment to encrypt and decrypt given sensitive data Use cryptography library fernet.

AIM:

To Conduct an experiment to encrypt and decrypt given sensitive data Use the cryptography library fernet.

PROGRAM:

```
from cryptography.fernet import Fernet
key = Fernet.generate_key()
cipher = Fernet(key)
data = "My password is 12345"
encrypted_data = cipher.encrypt(data.encode())
decrypted_data = cipher.decrypt(encrypted_data).decode()
print("Original Data:")
print(data)
print("\nEncrypted Data:")
print(encrypted_data)
print("\nDecrypted Data:")
print(decrypted_data)
```

OUTPUT:

```
Original Data:
My password is 12345

Encrypted Data:
b'gAAAAABqadsmPyzIuKKli9XeWhRJgqQIHXtcvkbCOgnQnpGRoqPh4k28pqIJyGW
IFfREL4D0OV0dLFV9sYMQfGn2HZvEe3RFIwgzImlrmG2P7y-LM3TbRx0='

Decrypted Data:
My password is 12345
```

RESULT:

Thus the program to Conduct an experiment to encrypt and decrypt given sensitive data Use the cryptography library fernet.